

The empirical recurrence for OEIS A231520, recovered from its tail

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Abstract

OEIS A231520 records “Empirical recurrence of order 70” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 103$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{70} = 32768 \neq 0$, so no further tail satisfies anything shorter; any order-70 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A231520 is “Number of nX5 0..1 arrays with no element less than a strict majority of its horizontal, diagonal and antidiagonal neighbors.”. It has offset 1 and begins

12, 342, 4665, 70462, 1031105, 15000701, 219937967, 3222629836, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:33:35, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 103$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 103$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 70: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 70.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 206$ exact terms from $n = 2$ on, it returns order 70 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	11	c_{19}	24387722085	c_{37}	55507891038645	c_{55}	522132318132
c_2	12	c_{20}	1201639640	c_{38}	−59442065196896	c_{56}	−38051064857
c_3	751	c_{21}	50099974247	c_{39}	69675601114071	c_{57}	−139073227125
c_4	−1532	c_{22}	−183156967699	c_{40}	−78748922952837	c_{58}	30095536745
c_5	−2267	c_{23}	118387574444	c_{41}	79090929279258	c_{59}	−19283803068
c_6	−104967	c_{24}	−377595765307	c_{42}	−81126387410676	c_{60}	18581955258
c_7	32494	c_{25}	1049801384207	c_{43}	85610068984871	c_{61}	18285137578
c_8	737259	c_{26}	−973790814339	c_{44}	−76834471355568	c_{62}	−8882546912
c_9	5194829	c_{27}	2085275289192	c_{45}	70570211671968	c_{63}	−261179976
c_{10}	−4230634	c_{28}	−4432674071130	c_{46}	−65507615914544	c_{64}	−952968056
c_{11}	−40705286	c_{29}	4432995271917	c_{47}	51791266712627	c_{65}	−279182592
c_{12}	−69605540	c_{30}	−8196934017000	c_{48}	−41676440967191	c_{66}	3471520
c_{13}	161293677	c_{31}	13606061306082	c_{49}	31992422233922	c_{67}	3626496
c_{14}	539489142	c_{32}	−14041808549101	c_{50}	−21438759993289	c_{68}	−4608256
c_{15}	591682632	c_{33}	22865128297655	c_{51}	13959058322318	c_{69}	2893824
c_{16}	−2444906181	c_{34}	−31157881032319	c_{52}	−8310563285898	c_{70}	32768
c_{17}	−3081409946	c_{35}	33286757819682	c_{53}	4054011737519		
c_{18}	−5381530648	c_{36}	−46080329722084	c_{54}	−1508089900711		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A231520 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{70} - c_1 x^{69} - \dots - c_{70}$. Its constant term is $-c_{70} = -32768$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 70 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 70, so $\deg q = 70$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 70, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 70, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 32768.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-70 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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